5G-SA Band-Combination Test Strategy

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1. Objective

Create a repeatable, measurable test suite that validates **5G Stand-Alone (5G-SA)** operation across **multiple NR band combinations** (e.g., N78 + N41, N78 + N79, N78 + N258, etc.). The suite must prove that the device under test (DUT) can:

1. **Select, lock-on, and maintain radio links** on each configured band.
2. **Perform inter-band handover** (both intra-RAT and inter-RAT) without service interruption.
3. **Satisfy key performance indicators (KPIs)** for throughput, latency, and reliability per 3GPP Release 18 specifications.

The outcome is a **pass/fail verdict** for each band-combination scenario that can be reported to product management and regulatory compliance teams.

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2. Scope

In-Scope	Out-of-Scope
• 5G-SA operation on NR bands N78, N41, N79, N258, N260, N261 (selected per market) • Dual-band (or multi-band) simultaneous transmission/reception • Mobility scenarios: static, pedestrian (3km/h), vehicular (30km/h) • Core-network interaction (SMF/UPF)	• NSA (EN-DC) operation • mmWave bands beyond N261 (e.g., N277) • Non-radio functional testing (UI, power-off behavior) • Certification-level RF compliance (SAR, emissions)

for data plane KPI measurement	

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3. Test Approach

3.1 Test Levels

Level	Description	Primary Artifacts
Lab Functional	Verify basic attach, registration, and service continuity on each band individually.	Test scripts, log captures, PASS/FAIL matrix
Integration (Dual-Band)	Simultaneously activate two configured bands; verify carrier aggregation (CA) or dual-connect behavior.	KPI dashboards, handover logs
Mobility (Inter-Band Handover)	Drive the DUT through coverage zones that force handover from BandA → BandB (and reverse).	Time-to-handover, packet loss, latency spikes
Performance Stress	Apply maximum-throughput traffic (e.g., 2Gbps UDP) on each band to verify sustained KPI compliance.	Throughput graphs, error-rate counters

3.2 Test Methodology

1.

Define Band Matrix – Create a **Band-Combination Matrix (BCM)** that lists all pairwise (or multi-band) permutations to be exercised.

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Configure RF Test Environment – Use a 5G-SA base-station emulator (e.g., Keysight 5G Test Set) capable of generating the required bands, with programmable path loss to emulate cell edge / indoor conditions.

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Automate Test Execution – Develop **Python-based test harness** that:

6.

- Sends AT/SSH commands to the DUT for attach/detach.
- Controls the emulator (band on/off, power level, handover trigger).
- Collects KPI data via **IPERF3**, **ping**, and **RAN-level counters** (RRC, PDCP).

7.

Apply Pass/Fail Criteria – For each KPI, set thresholds derived from 3GPP TS 38.104/38.306 and product specifications (see Section 5).

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Report & Traceability – Map each test case to **Requirement IDs** (e.g., R-5G-SA-B01) and capture result in **Test Management Tool** (e.g., Zephyr, TestRail).

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4. Test Cases – Step-by-Step Generation

Below is a **template** that the test architect can replicate for every band-combination entry in the BCM.

4.1 Template

Field	Description
TC-ID	Unique identifier (e.g., TC-5GSA-BC-N78-N41-001)
Title	*"Verify 5G-SA dual-band operation on N78 + N41 – Static"*
Related Requirement(s)	R-5G-SA-B01, R-5G-SA-B02
Pre-conditions	• DUT powered on, factory-reset • SIM provisioned for 5G-SA • Emulator configured with N78 (3.5GHz) and N41 (2.5GHz) carriers, each with 20MHz BW, TxPower = 23dBm • DUT set to "Auto-Band-Selection"

Test Steps	1. Initialize – Verify DUT is in idle state (no RRC connection). 2. Activate BandN78 – Enable N78 carrier on emulator, set path loss = 70dB (cell-edge). 3. Attach – Issue attach command; capture RRC Connection Setup time. 4. Validate KPI – Run IPERF3 UDP 1Gbps for 30s; record throughput, jitter, packet loss. 5. Activate BandN41 – Enable N41 carrier while N78 remains active. 6. Trigger Dual-Band – Issue “Enable Carrier Aggregation” command (if applicable) or let DUT automatically add N41. 7. Measure Aggregated Throughput – Run IPERF3 again; verify combined throughput ≥ 1.8Gbps. 8. Stability Check – Keep both carriers active for 5min; monitor RRC re-configuration events. 9. Detach – Issue detach; verify clean release of both carriers.
Expected Result	• RRC Connection Setup ≤ 150ms • Single-band throughput ≥ 800Mbps (±5%) • Dual-band aggregated throughput ≥ 1.8Gbps (±5%) • Packet loss ≤ 0.5% • No RRC drops during 5-min stability period
Pass/Fail Criteria	PASS if all KPI thresholds met; otherwise FAIL (record which KPI failed).
Post-conditions	DUT returns to idle; emulator carriers disabled.
Attachments	Log files (RRC dump, iperf3 output), screenshots of KPI dashboard.

4.2 Example Instantiation – N78 + N258 (mmWave)

Field	Value
TC-ID	TC-5GSA-BC-N78-N258-001
Title	Verify 5G-SA dual-band operation on N78 (3.5GHz) & N258 (26GHz) – Pedestrian Mobility
Pre-conditions	Same as template, plus: • N258 carrier with 100MHz BW, TxPower=20dBm • Path loss N258 = 80dB (simulated outdoor)
Test Steps	1-4: Same as template on N78. 5-6: Enable N258; command DUT to “Add Secondary Cell”. 7. Mobility – Move DUT (robotic stage) at 3km/h through a zone where N258 signal degrades (increase path loss from 80dB to 95dB). 8. Verify Seamless Handover to N78-only without >50ms latency spike.
Expected	• Dual-band throughput ≥1.5Gbps (N78+N258) •

Result	Handover latency ≤30ms • No packet loss >1% during handover
**Pass	